

Microcontroller-based Bit Error Rate Test set

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1 Project Description

Communication networks are continuously expanding in capacity and speed and being able to ensure data integrity is paramount. The most common example of such a system is the internet, which uses undersea optical fibres for data transmission. (Ramble about the impact of the internet here) Channels for digital communication are susceptible to incorrect data transmission arising due to many possible factors such as noise or interference. A metric of performance for digital communication channels is the Bit Error Rate (BER) which can be characterised via a Bit Error Rate Test (BERT). [1] This project aims to construct a system that can (describe project)

2 Literature Review

References

- [1] S. M. J. Alam, M. R. Alam, G. Hu, and M. Z. Mehrab, “Bit error rate optimization in fiber optic communications,” *International Journal of Machine Learning and Computing*, vol. 1, no. 5, pp. 435–440, Dec 2011, doi:<http://dx.doi.org/10.7763/IJMLC.2011.V1.65>.